THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 8: **Heredity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Traits that a parent acquires during life, such as a bodybuilder's muscle, are 1

(a) transmitted by DNA **(b)** inherited always
(c) not inherited **(d)** inherited only by sons

2. The trait that is not seen in F1 but reappears in F2 is 1

(a) sex-linked **(b)** dominant **(c)** recessive **(d)** acquired

3. A homozygous plant is 1

(a) always tall **(b)** Tt **(c)** always short **(d)** TT or tt

4. Which of the following is an inherited trait? 1

(a) Eye colour **(b)** Tanned skin
(c) Learning to cycle **(d)** A scar on the hand

5. Alternate forms of a gene are called 1

(a) chromosomes **(b)** zygotes **(c)** gametes **(d)** alleles

6. A cross between F1 (Tt) and the recessive parent (tt) is called 1

(a) reciprocal cross **(b)** a test cross **(c)** self-cross **(d)** a dihybrid cross

7. Free earlobes and attached earlobes are examples of 1

(a) acquired traits **(b)** variation within a species
(c) sex determination **(d)** no variation

8. The physical appearance of an organism for a trait is its 1

(a) genome **(b)** genotype **(c)** phenotype **(d)** allele

9. The gametes of a Tt plant are 1

(a) only Tt **(b)** T and t in equal numbers
(c) only T **(d)** only t

10. Which gamete decides that the baby is a girl? 1

(a) Sperm with Y chromosome **(b)** Egg with O
(c) Sperm with X chromosome **(d)** Egg with Y

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the F2 generation show the F1 hidden trait again? 2

12. Differentiate between genotype and phenotype. 2

13. How is the sex of a child determined? 2

14. What is a gene? Where is it present? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with a cross why a tall F1 plant may produce dwarf offspring. 3

16. In pea plants, round seeds (R) are dominant to wrinkled (r). What is the result of crossing Rr x rr? 3

17. How does the Y chromosome influence the sex of a baby? 3

OR

Explain sex determination in humans with a chart. 3

18. Why is variation more in sexual reproduction? 3

OR

State the difference between a test cross and a self-cross. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between acquired and inherited traits with two examples each. Why do acquired traits not pass on? 5

OR

Explain why a child may resemble the parents but also differ from them. 5

20. How do traits get expressed? Explain the link between genes, proteins and traits with the example of plant height. 5

OR

Describe the different ways in which variations arise in sexually reproducing organisms. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND